

Name: _____

Date: _____

STRAIGHT-LINE GRAPHS, (PLOTting STRAIGHT-LINE GRAPHS)

LO: I can plot straight-line graphs by generating a table of values and drawing the line.

Plotting straight-line graphs

To **plot a straight line**, substitute values of x into the equation to find corresponding y -values. Plot these **coordinate pairs** and join with a ruler.

Tip: You only need two points to draw a line, but use three to check for errors.

Quick examples

●	$y = 2x + 1$: when $x = 0$, $y = 1$; when $x = 2$, $y = 5$	substitute x values
●	Use $x = 0, 1, 2, 3$ to make a table of values	systematic approach
●	Three collinear points confirms the line	error check
●	Only need one point to draw a line	need at least two

Key idea

Substitute at least three x -values into the equation. Plot the (x, y) pairs and join with a straight line.

$$x = 0 \quad y = \textcircled{c}, \quad x = 1 \quad y = \textcircled{m + c}$$

Start with $x = 0$ to find the y -intercept

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - table of values

Complete a table for $y = 3x - 1$ with $x = 0, 1, 2, 3$.

$$x = 0: y = -1; \quad x = 1: y = 2$$

$$x = 2: y = 5; \quad x = 3: y = 8$$

Points: $(0, -1), (1, 2), (2, 5), (3, 8)$

y values: -1, 2, 5, 8

Substitute each x -value into the equation.

Example 2 - using intercepts

Plot $y = 2x - 4$ using x and y intercepts.

y -intercept: $x = 0$, $y = -4$ gives $(0, -4)$

x -intercept: $y = 0$, $2x = 4$, $x = 2$ gives $(2, 0)$

Plot $(0, -4)$ and $(2, 0)$ and join

Line through $(0, -4)$ and $(2, 0)$

Intercepts give two quick points without a full table.

Example 3 - negative gradient

Plot $y = -x + 3$ for $x = 0$ to 4 .

$$x = 0: y = 3; \quad x = 1: y = 2; \quad x = 2: y = 1$$

$$x = 3: y = 0; \quad x = 4: y = -1$$

Line slopes downward from $(0, 3)$ to $(4, -1)$

Downward sloping line

A negative gradient makes the line slope down from left to right.

Example 4 - fractional gradient

Plot $y = \frac{1}{2}x + 1$ for $x = 0, 2, 4, 6$.

$$x = 0: y = 1; \quad x = 2: y = 2$$

$$x = 4: y = 3; \quad x = 6: y = 4$$

Gentle slope, rising by 1 for every 2 across

Points: $(0, 1), (2, 2), (4, 3), (6, 4)$

Choose even x -values to avoid fractions in the y -coordinates.

YOUR TURN – PRACTICE (deliberate practice)

1. Completing tables

Complete the table of values for $x = 0, 1, 2, 3$.

- a) $y = 2x + 3$
- b) $y = x - 2$
- c) $y = 4x - 1$
- d) $y = -2x + 5$

2. Finding intercepts

Find the x-intercept and y-intercept.

- a) $y = 3x - 6$
- b) $y = -2x + 8$
- c) $y = x + 4$
- d) $y = 5x - 10$

3. Negative and fractional gradients

Complete values for $x = -2, 0, 2, 4$.

- a) $y = -3x + 2$
- b) $y = \frac{1}{2}x + 3$
- c) $y = -\frac{1}{2}x + 4$
- d) $y = \frac{3}{2}x - 1$

4. Identifying equations from points

Which equation matches the points given?

- a) (0, 2) and (1, 5)
- b) (0, -1) and (2, 3)
- c) (0, 4) and (2, 0)
- d) (0, 0) and (3, 6)

5. Mixed plotting problems

Solve each problem.

- a) $y = x + 3$. Value of y when $x = -2$? b) $y = 2x - 5$. When does $y = 0$?
- c) Lines $y = x + 1$ and $y = 3x - 3$. Where do they meet?
- d) $y = -x + 6$. Value of x when $y = 2$?
- e) $y = 4x - 8$. Give 3 points on this line.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) You need at least two points to draw a straight line ☐

Correct answer: _____

Reason: _____

b) You only need one point to plot a line ☐

Correct answer: _____

Reason: _____

c) The y-intercept is found by setting $x = 0$ ☐

Correct answer: _____

Reason: _____

d) For $y = \frac{1}{2}x + 1$, when $x = 1$, $y = 1.5$ ☐

Correct answer: _____

Reason: _____

e) All straight lines pass through the origin ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Plot the lines $y = 2x - 1$ and $y = -x + 5$ for $x = 0$ to 4. Find the coordinates of their point of intersection by solving simultaneously.

Expression: _____

Simplified: _____

b) A taxi company charges a **£2.50 base fare** plus **£1.20 per km**. Write an equation for cost (C) in terms of distance (d). Plot the graph for 0 to 10 km. Use your graph to estimate the cost for **7.5 km**.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

